

versus them’ tendencies. Additionally, persona creation was implemented through prompt-based conditioning rather than fine-tuning. Future work could focus on fine-tuning LLMs with persona-specific datasets to explicitly control persona representation, on better understanding how training data composition influences social, psychological, and cognitive biases, or on how those biases are enacted under what conditions (among other options). More specifically, future work could look at the relationship between training objective(s) and the expression of psychological biases. For example, do global cognitive biases show up in pure next-token-prediction models, or only once pragmatics are more explicitly invoked as in instruction-tuning? We are interested whether the larger pragmatic goals could be more amenable to learning distributed patterns that are potentially only meaningful in aggregate (which may or may not be the case for the expression of any known psychological biases in language); for more discussion of related topics, see [Zimmerman \(2025\)](#); [Zimmerman et al. \(2025\)](#). Future work could also explore how bias mitigation strategies, including the one proposed here, impact other aspects of the output, such as the topics discussed and the style it is written in.

6.0.1 Normative stance

While we do not take a normative stance as to whether or not models should reproduce which human biases under what circumstances — not because it is not important, but because it deserves to be the focus of its own project(s), rather than an addendum to this one — we do think it is important and useful to understand the existing terrain. That is, it is important to understand how LLMs enact the biases they absorb through their training data under what conditions (e.g. user-generated context), how their architecture (including the structure of their training regimes) inculcates biases, and how those two pieces interact, especially as LLMs become more ubiquitous. This information is necessary for their responsible deployment. We hope that by describing what in-group and out-group bias look like in LLMs, and how that bias can be mitigated at the output level in the case where that is appropriate, we are contributing to that goal. For psychological biases in particular, they may be contextually appropriate, and valuable as heuristics, so whether they should be reflected in LLM output depends on the application of that output. More generally, when it comes to cognition, there is no

view from nowhere; there must always be some bias (“bias” as in, skew from what is being represented, not as in “social bias”); more discussion of this topic in [Zimmerman \(2025\)](#). Whether any aspect of what might broadly be called ‘us versus them’ bias is innate to having a perspective, a self, is out of scope for this project, but not obviously implausible.

6.0.2 ‘Us versus them’ bias simplification

Related to the above limitation (Sec. 6.0.1), there is a wide range of human behavior which could fall under the umbrella of ‘us versus them’ bias, depending on context. We are considering a narrow subset: ingroup solidarity and outgroup hostility as expressed through sentences starting with “We are...” or “They are...”, and as measured via our specific methodologies. This is a very circumscribed lens through which to examine bias. Bias is a complex topic, especially when it comes to a psychological bias like this one, which manifests as a wide array of behaviors, including as various social biases.

6.0.3 Doubly-terminated fallacy of interpretation: Theory of Mind and language as beguiling medium

It is important to note that in using psychological and behavioral concepts developed for people to talk about LLMs, we are exchanging correctness for parsimony. We are discussing LLM behaviour, that is, text output, as if it were language output by a human: If a human wrote such a sentence, we would plausibly interpret that sentence as exhibiting ‘us versus them’ bias. We elide that to: the LLM exhibits ‘us versus them’ bias. We do not actually know what relationship that apparent bias has to most of the LLM’s internal processes, other than that it is ultimately produced by them; we do not know if the LLM “feels” biased. We also do not know if that apparent bias goes along with any of the things it might be expected to go along with for people. In other words, the familiar Theory of Mind applies from neither direction, despite the temptation of human-like text output ([Zimmerman, 2025](#)), making construct validity an open question.

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